

Ghost Cache

LoRa sneakernet without the sneakers

Delay-tolerant content replication over infrastructureless radio

A ~~~~~> B ~~~~~> C

What if the Internet is not available, but the information still needs to move?

INTERNET

Fast—when it exists

Unavailable, unreliable, damaged, or restricted.

USB HANDOFF

Reliable—but manual

Every update requires another physical trip.

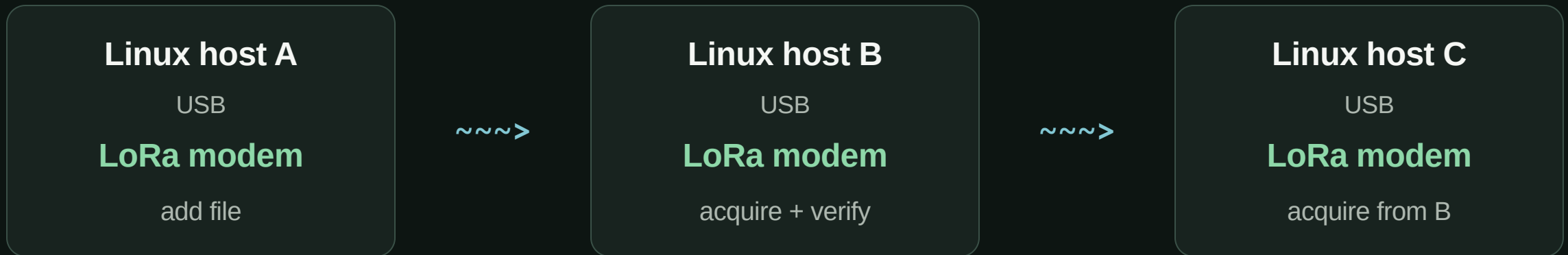
LORA

Tiny bandwidth—autonomous

Compact information can propagate after deployment.

Ghost Cache does not replace broadband. It targets compact information when infrastructure is absent or intermittent.

Add once. Verify. Relay onward.



Once a node verifies content, it becomes another source.

No Wi-Fi · no cellular · no Internet · no central server · original source may disappear

Airtime is the scarce resource

EXAMPLE US RF PROFILE

915 MHz US example

SF7 · 125 kHz · CR 4/5 · 5 dBm

A low-power example—not a universal deployment recommendation.

THE TRADEOFF

faster
less airtime



more sensitivity
more airtime

lower SF

higher SF

4 KiB ≈ 12 sec

small bulletin

16 KiB ≈ 46 sec

demo-sized publication

64 KiB ≈ 3.1 min

larger reference pack

Rule of thumb: ≈ 3 sec/KiB. Idealized clean-link transfer phase only; discovery, metadata, loss, retries, contention, and slower RF settings add time.

The network tracks content, not routes

Publication = one file plus immutable metadata: hash, filename, chunk layout, and optional signature.

DISCOVER

INSPECT

DECIDE

TRANSFER

VERIFY

REPLICATE

RECEIVER-DRIVEN

A source advertises what exists. Each receiver decides what it will acquire.

DELAY-TOLERANT

Partial transfers persist, resume after restart, and can continue from another source.

Compare differences, not entire inventories

PLAIN-ENGLISH GOAL

Node A: A B C D E

Node B: A B C D

Discover only:

"B is missing E."

RADIX-PREFIX DIGEST RECONCILIATION

Compare compact summaries.

Descend only through differing ID prefixes.

Send concrete IDs only at small leaves.

5 packets · 484 bytes

1 difference among 1,000 objects

7 packets · 686 bytes

1 difference among 10,000 objects

Software measurement: cost follows differing branches—not every publication ID.

Trust belongs to the receiver

SHA-256

Are these the exact bytes named by the manifest?

Ed25519

Did this signing key sign this publication metadata?

Trust policy

Does this receiver accept that key—or require any signature at all?

A signs X

B relays unchanged

C trusts A

C verifies A—even when X arrives from B

want = explicit local override

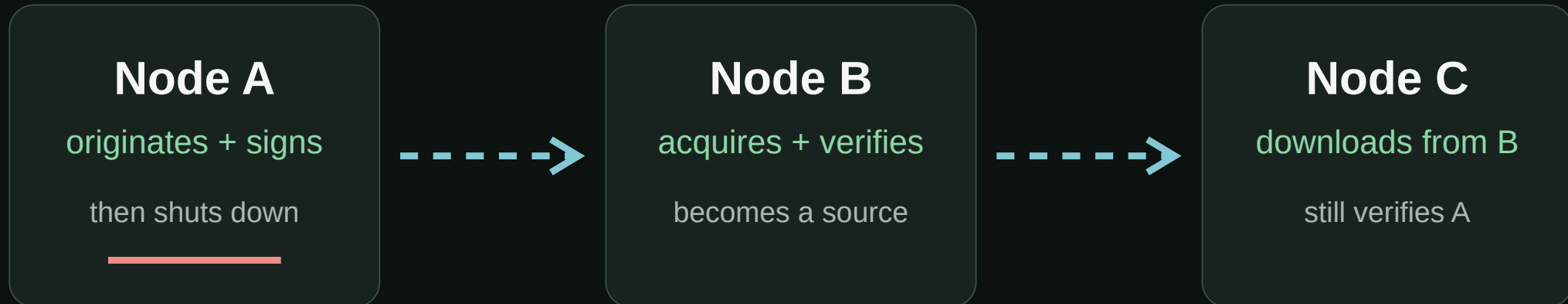
May override unsigned / untrusted policy

Never overrides cryptographic failure

Traffic is plaintext. Signing is not encryption, anonymity, or identity bootstrap.

Demo 1 · The source disappears

RELAY DEMO



A signs → B acquires → A stops → C acquires from B → C verifies A

Proof points: A's key ID remains trusted at C · full SHA-256 matches · content bytes compare equal

PDF / missing-video fallback: A signs → B acquires → A stops → C acquires from B → C verifies A

Demo 2 · The receiver is in control

POLICY + RESUME DEMO



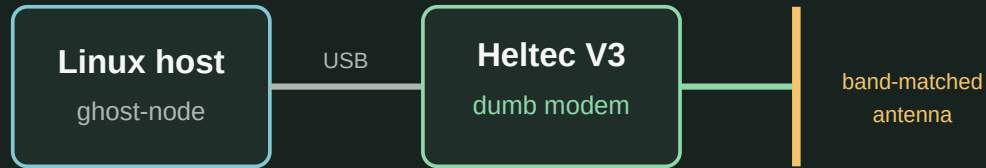
defer → inspect → approve live → interrupt → resume from durable chunks

The final SHA-256 must still match. A want never accepts corrupt content.

Proof points: partial chunk state survives · restart requests only missing chunks · final SHA-256 matches

PDF / missing-video fallback: defer → library → live want → interrupt → durable resume → verify

Headless deployment



Host owns replication • firmware owns raw radio TX/RX

DEPLOYMENT CAPABILITIES

- RF settings from TOML
- Apply + verify modem settings at startup
- Non-root systemd service
- Stable serial paths
- Volatile operational logs
- Persistent content + recovery state

Content and recovery state persist. Operational logs are volatile by default.

What Ghost Cache is—and is not

PHYSICALLY VERIFIED

- Three-node LoRa replication
- Arbitrary binary publications
- Interrupted resume + source switching
- Receiver-driven policy + live want
- Radix reconciliation
- Signed relay + trusted origin verification
- Host-applied RF configuration

CURRENT BOUNDARIES

- 247.5 KiB publication maximum
- Plaintext RF; no anonymity
- No freshness or revocation semantics
- No automatic retention or GC
- Heltec V3 reference modem
- Range is deployment-specific
- No general routed messaging

The sender can disappear. The information doesn't have to.

Ghost Cache

content-centric · receiver-driven · delay-tolerant

A ~~~~~> B ~~~~~> C

RF knobs in one slide

Term	Practical meaning
LoRa	Low-bandwidth, long-range-oriented packet radio modulation
LoRaWAN	Gateway/network-server protocol; not used by Ghost Cache
Frequency	Where radios communicate; must match and be locally legal
Spreading factor	Lower is faster; higher can improve sensitivity but costs airtime
Bandwidth	Wider is faster; narrower may improve sensitivity
Coding rate	Error-correction redundancy; stronger coding costs airtime
RSSI / SNR	Received strength / signal relative to noise; diagnostics, not guarantees

GN v3 message flow

A advertises radix SUMMARY
B requests only differing prefixes
A returns SUMMARY or concrete LEAF IDs

B → GET_MANIFEST A → MANIFEST
B → GET_SIGNATURE A → SIGNATURE

B applies local policy

B → GET_CHUNKS A → up to 4 CHUNK packets
... repeat missing chunks ...
B verifies SHA-256 + signature, then commits

200-byte application packets · one self-initiated addressed exchange at a time · randomized timing and bounded backoff

Publication format

IDENTITY + INTEGRITY

Publication ID

first 128 bits of content SHA-256

Authoritative hash

full SHA-256 retained and verified

TRANSFER + AUTHENTICITY

Chunk content

178 bytes per packet

Optional signature

Ed25519 over canonical immutable metadata

Maximum today: $1,424 \text{ chunks} \times 178 \text{ bytes} = 253,472 \text{ bytes} \approx 247.5 \text{ KiB}$

Signing and local policy

Receiver policy	Unsigned	Valid unknown signer	Valid trusted signer	Invalid signature
permissive	acquire	acquire	acquire	reject
signed	defer	acquire	acquire	reject
trusted	defer	defer	acquire	reject

want overrides unsigned/untrusted and size admission—but never invalid signatures or SHA-256 failure.

Key ID = first 128 bits of SHA-256(raw Ed25519 public key). Key ownership is established outside Ghost Cache.

Reconciliation cost: paging vs radix

Inventory	Full-ID paging estimate	Radix, one difference
1,000 objects	≈ 91 inventory packets	5 packets / 484 bytes
10,000 objects	≈ 910 inventory packets	7 packets / 686 bytes

Paging estimate: 11 full 128-bit IDs per packet. Radix figures: deterministic Ghost Cache software measurements. Both exclude manifest inspection and content transfer.

The win appears when libraries are large and mostly equal.